

Águabots — Rescue Simulation 2026 Bill of Materials (BOM)

Prepared for RoboCupJunior Rescue Simulation. Prices reflect direct cost to the team for the listed software/tools.

Category	Software/tool name	Description	Author	Price (USD)	Price note	Project role	Source / Evidence
Core Environment	Ubuntu 24.04 LTS	Base operating system used in the project Docker environment.	Canonical Ltd.	\$0,00	Free / open-source	Runtime and build environment	https://ubuntu.com/
Core Environment	Docker	Containerized development and execution environment for repeatable setup.	Docker, Inc.	\$0,00	Free plan / open-source engine used	Reproducible environment	https://www.docker.com/
Simulation	Webots R2023b	Robot simulator and controller API used to run the Rescue Simulation robot.	Cyberbotics Ltd.	\$0,00	Free / open-source	Simulation and robot control	https://cyberbotics.com/
Middleware	ROS 2 Jazzy	ROS 2 communication layer for mapping, vision, topics, visualization, and debugging.	Open Robotics / OSRF	\$0,00	Free / open-source	ROS 2 processing layer	https://docs.ros.org/en/jazzy/
Visualization	RViz2	Visualization tool used to inspect ROS 2 data, robot state, mapper outputs, and debug markers.	ROS 2 / Open Robotics	\$0,00	Free / open-source	Debugging and visualization	https://github.com/ros2/rviz
Mapping	SLAM Toolbox	ROS 2 mapping support used as part of the mapping/debugging infrastructure.	Open Navigation / Steve Macenski	\$0,00	Free / open-source	Mapping support	https://github.com/SteveMacenski/slam_toolbox
Inference	ONNX Runtime 1.23.2	Runtime used to execute exported ONNX neural-network models during victim and wall-token detection.	Microsoft	\$0,00	Free / open-source	AI inference	https://onnxruntime.ai/
Computer Vision	OpenCV	Computer vision library used for image processing, target geometry, polar unwrapping, HSV decoding, and debugging.	OpenCV Team / OpenCV.org	\$0,00	Free / open-source	Vision processing	https://opencv.org/
Machine Learning	PyTorch	Machine-learning framework used to train the victim and wall-token neural-network models.	PyTorch Foundation	\$0,00	Free / open-source	Model training	https://pytorch.org/
Machine Learning	Google Colab	Cloud notebook environment used for model training and experimentation with available GPU resources.	Google	\$0,00	Free tier used	Training and experimentation	https://colab.research.google.com/
Programming	Python 3	Language used for data generation, notebooks, training scripts, experiments, and tooling.	Python Software Foundation	\$0,00	Free / open-source	Training and scripting	https://www.python.org/
Programming	C++17 / GCC / G++	Language and compiler toolchain used for the Webots controller and ROS 2 C++ nodes.	GNU Project / Free Software Foundation	\$0,00	Free / open-source	Controller and ROS 2 nodes	https://gcc.gnu.org/
Build Tools	CMake	Build configuration tool used by the C++ controller libraries and ROS 2 packages.	Kitware	\$0,00	Free / open-source	Build system	https://cmake.org/
Build Tools	colcon	ROS 2 workspace build tool used to compile project packages.	ROS 2 / Open Robotics	\$0,00	Free / open-source	ROS 2 build tool	https://colcon.readthedocs.io/
Data Processing	NumPy	Python numerical library used in image processing, dataset processing, and experiments.	NumPy Developers	\$0,00	Free / open-source	Training and analysis support	https://numpy.org/
Development	Visual Studio Code	Code editor used for programming, debugging, and project organization.	Microsoft	\$0,00	Free	Development environment	https://code.visualstudio.com/
Development	Git	Version-control system used to manage source-code history.	Software Freedom Conservancy	\$0,00	Free / open-source	Version control	https://git-scm.com/
Development	GitHub	Repository hosting platform used to store, manage, and share the project source code.	GitHub, Inc.	\$0,00	Free plan used	Repository hosting	https://github.com/
Custom Software	exploration_legacy	Custom controller-side navigation library for tile exploration, wall detection, hole handling, backtracking, and wheel-position movement.	Águabots Team	\$0,00	Team-developed	Final navigation	Team source code: controller/exploration_legacy
Custom Software	ROS 2 Mapper	Custom ROS 2 mapper package that builds a semantic tile map from odometry, LiDAR scans, floor classification, and map data.	Águabots Team	\$0,00	Team-developed	Semantic mapping and final matrix support	Team source code: controller/ros2_ws/src/ros2_mapper
Custom Software	ROS 2 Vision	Custom ROS 2 vision package that receives camera frames through TCP and runs ONNX/OpenCV detection pipelines.	Águabots Team	\$0,00	Team-developed	Victim and wall-token detection	Team source code: controller/ros2_ws/src/ros2_vision
Custom Software	TCP Bridge	Custom TCP communication layer between Webots and ROS 2 for sensor data, state, frames, and result messages.	Águabots Team	\$0,00	Team-developed	System integration	Team source code: controller/ros2_ws/src/ros2_bridge
Custom Software	ONNX Detection Models	Team-trained ONNX models for object presence, object type, victim bounding-box localization, and victim classification.	Águabots Team	\$0,00	Team-developed	AI perception models	Team source code: controller/models and notebooks

Total direct software/tool cost to the team

\$0,00 All listed items were used at zero direct cost via open-source licenses, free tools, or team-developed software.